Individual differences

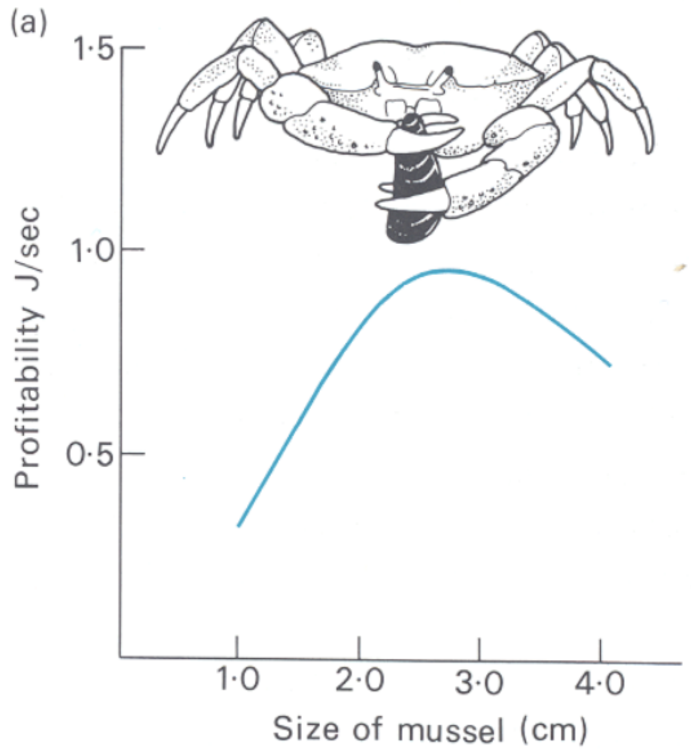
IB 431 – Behavioral Ecology
Fall 2025
September 25th 2025

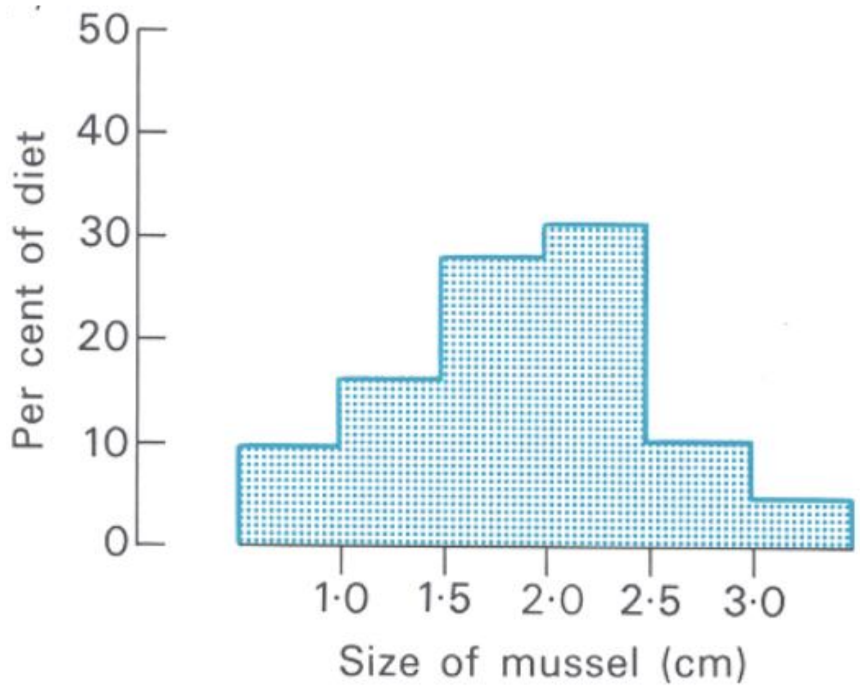
Outline

I. Behavioral types, plasticity, and individual differences
II. Drivers of behavioral types
III. Discuss Laskowski et al 2022

Individual differences in behavior

- Behavioral ecology historically driven by optimality theory
- Focus on mean trait value – variation is just noise





Individual differences in behavior

- Distinct and repeatable variation in behavior among individuals of a species
  - Behavioral types
  - Temperament
  - Personality

Behavioral types

- Exploration
- Aggression
- Risk-taking / boldness
- Foraging strategies
- Social tendencies (i.e., more solitary vs. connected)
- Parental care
- Courtship

Bold-shy continuum in pumpkinseed sunfish

- Boldness measured multiple ways
  - Response to a novel, potentially threatening stimulus
  - Response to a known predator
  - Willingness to forage in open water

Wilson et al 1993

Behavioral types

- Communication modes
  - Blue whales may differ in their calls and acoustic signals (Stafford et al 2001; Lewis et al 2018)
  - Typical frequency: 10-39 Hz

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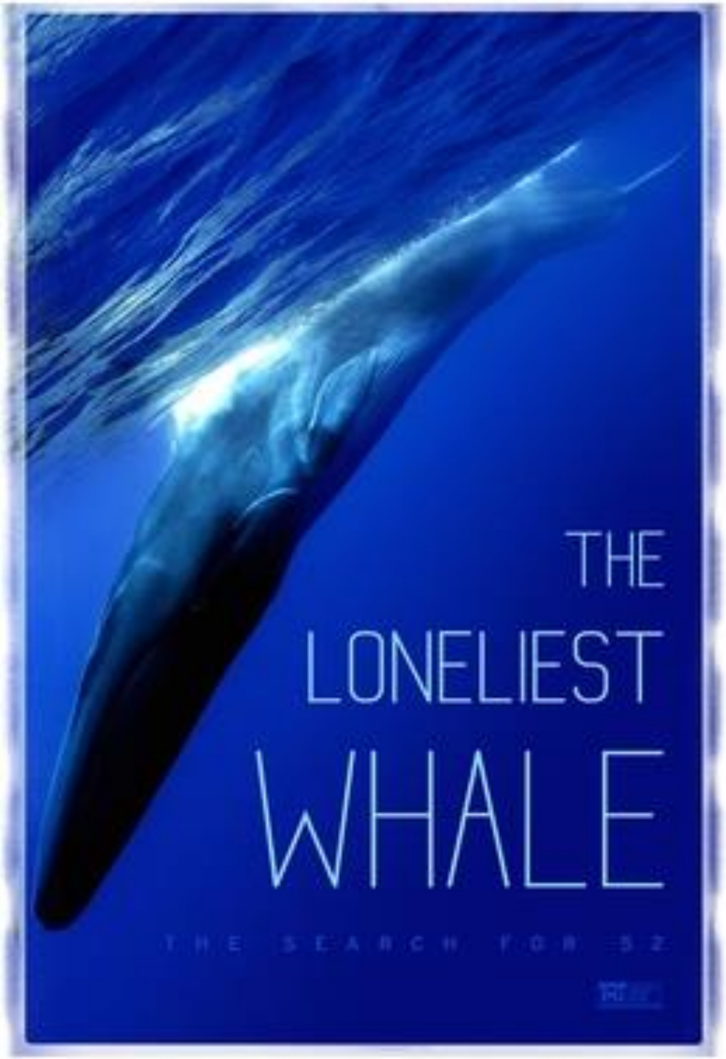
- Communication modes
  - Blue whales may differ in their calls and acoustic signals (Stafford et al 2001; Lewis et al 2018)
  - Typical frequency: 10-39 Hz
  - 52 Hertz whale – individual detected calling at the wrong frequency



Behavioral syndromes

- Correlation between multiple behavioral traits
  - Exploration-aggression
  - Boldness-aggression
  - Territorial behavior and antipredator behavior in skinks (Stapley & Keogh 2005)

Carryover effects

- Behavioral types can may have unintended consequences for other functions
  - Aggressive behavior develops to increase foraging rate in fishing spiders
  - During reproduction, increased aggression causes females to eat males (sexual cannibalism)

Johnson 2001

Carryover effects

Negative frequency-dependent selection

- When less common behaviors have increased fitness
- Can maintain multiple behavioral types
  - Marine isopod mating strategies

Schuster 1992

Behavioral type vs. plasticity

- Behavioral type: individuals differ consistently in a behavioral trait (e.g.; some more aggressive, others less)

- Plasticity: individuals can switch between different behaviors (e.g., aggressive in some conditions, less aggressive in others)

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Why might individuals differ in behavioral type?

- **Variation in other phenotypic traits (e.g., morphology, physiology)**

Morphological and physiological influences

- Coho salmon – body size leads to different strategies
  - Hooknoses: larger, slower development time, more aggressive in courtship, compete directly with other males
  - Jacks: smaller, faster development, "sneaky" mating strategy

Tanaka et al 2009

Why might individuals differ in behavioral type?

- Variation in other phenotypic traits (e.g., morphology, physiology)
- **Developmental conditions**

Development of behavioral types in pigs

- High (HR) vs. low resisting (LR) pigs (measured using the "backtest")
- Reared in barren or enriched conditions
- Barren-housed = less active, less play behavior, less social
- Enriched conditions were particularly impactful on LR pigs

Bolhuis et al 2005

Why might individuals differ in behavioral type?

- Variation in other phenotypic traits (e.g., morphology, physiology)
- Developmental conditions
- **Variation in cognitive abilities**

Cognition and behavioral types

- Individual differences in learning / brain functioning can drive behavioral differences
- Learning in red junglefowl was predictive of exploratory behavior
  - Age-dependent (for chicks more explorative were better learners, opposite for adults)
  - Sex-dependent (females learned faster)

Zidar et al 2018

Why might individuals differ in behavioral type?

- Variation in other phenotypic traits (e.g., morphology, physiology)
- Developmental conditions
- Variation in cognitive abilities
- **Life history trade-offs**

Life history trade-offs

- Pace of life syndrome (similar to r vs. K selection)
  - Fast vs. slow development time, metabolic rate, reproduction timing
- Life history strategy often shapes risk-taking or exploration
  - Faster pace of life = might favor increased risk-taking, exploration

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- Fairy wrens – more active, aggressive birds died sooner

Hall et al 2015

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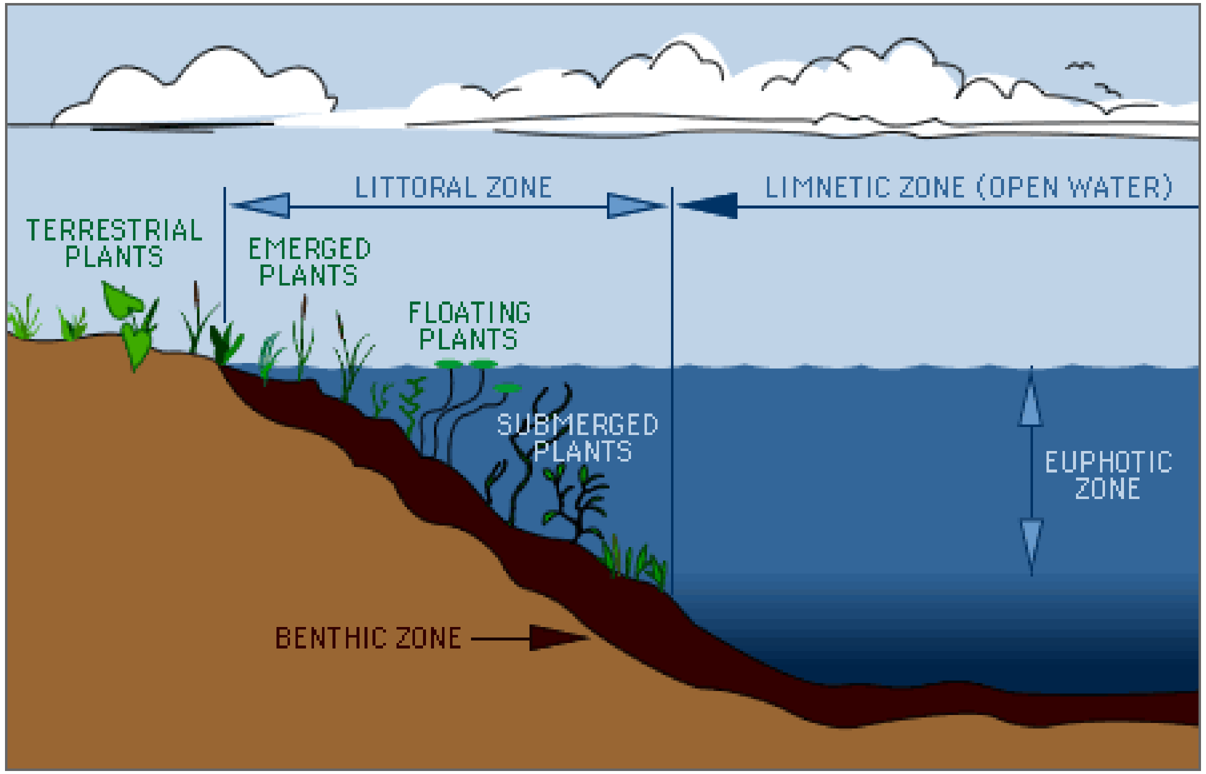
- Variation in other phenotypic traits (e.g., morphology, physiology)
- Developmental conditions
- Variation in cognitive abilities
- Life history trade-offs
- **Habitat use / microenvironmental differences**

Microenvironmental drivers of behavior

- There may be fine-scale differences within a species' habitat
- Benthic and limnetic ecotypes in three-spined stickleback

Foster et al 2008

Benthic and limnetic stickleback



Foster et al 2008

Benthic and limnetic stickleback

- Benthic = more plant cover, diet of small invertebrates, crustaceans, etc.
- Limnetic = open water, zooplankton diet
- Differences in courtship styles, boldness, shoaling behavior, foraging strategies

Foster et al 2008

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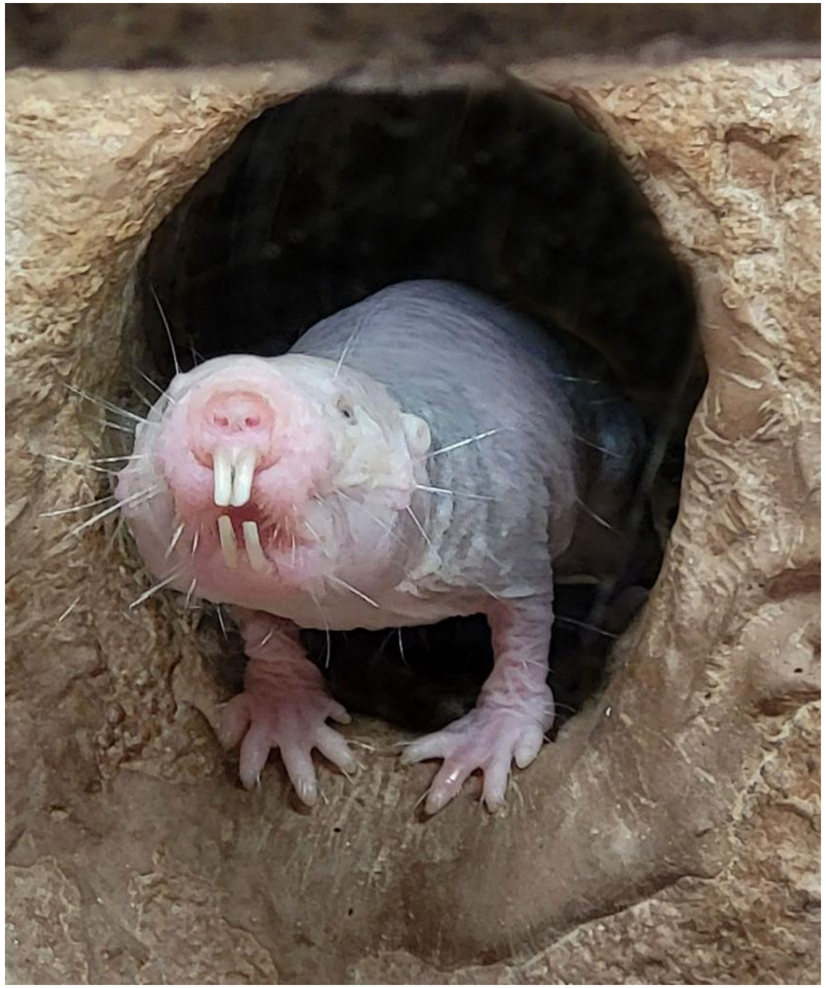
- Variation in other phenotypic traits (e.g., morphology, physiology)
- Developmental conditions
- Variation in cognitive abilities
- Life history trade-offs
- Habitat use / microenvironmental differences
- **Social environment**

Social niche specialization hypothesis

- Predicts that in a social group, individuals will take on partitioned, specialized roles to reduce conflict
  - E.g., “soldiers” vs. “foragers”
- Most prominent example – social insects (ants, bees, wasps, termites)
- Evidence in other species is limited



Can behavioral types emerge in the absence of environmental variation?

- Genetic differences can lead to intrinsic differences in behavior
- What about in asexual / clonal species?

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